

Private Acoustic Telemetry Coverage Areas

Type Enterprise Geodatabase Feature Class
Tags RWSC, Acoustic Telemetry, Fish, Receivers, ACT, FACT

Summary

Acoustic telemetry has revealed a great deal about species movement that informs fisheries management. In the U.S. this method is increasingly being used as a cost-effective way to gather data on species from rivers to oceans. Displaying receiver locations on a research planning map can benefit stakeholders from every sector and allow for the informed planning of asset placement that leverages the existing network coverage. A few benefits include: reducing the inadvertent expensive duplication of receiver coverage, conducting better science as more and better-placed receivers lead to higher potential statistical power and certainty in results, and providing more context for decision makers. While the Research Planning map is not designed to serve as a navigational safety tool, it can also increase awareness of receiver presence by other ocean users, which can reduce conflict and increase longevity of deployments. While acoustic Telemetry is a data collection method used for offshore wind research and monitoring, this method has been around for decades to study animal movement and behavior and is widely used for other purposes.

Description

This dataset displays bounding boxes depicting the locations of private acoustic telemetry receiver coverage areas submitted to the Regional Wildlife Science Collaborative Protected Fish Species Subcommittee. The layer only includes projects that have receivers deployed beginning on 01/01/2022, including current projects and planned projects whose receivers have not yet been deployed. Bounding boxes are provided by project personnel and plotted using the most recent deployment locations provided. The project/deployment start dates are also noted, and it can be assumed that receivers have been deployed throughout the duration of the project. Note that receiver deployments may change over the life of a project, and stations may go in and out of use. If you would like more specific information about a project's deployments, please reach out to the project personnel using the provided contact information. Bounding boxes are symbolized by color according to the status of the project at the time of the last update. The color of the receiver deployments will not adjust to the time slider and will always depict the current project status.

Credits

RWSC Protected Fish Subcommittee: Jordan Katz, jordan.katz@noaa.gov

Use limitations

These data will be used by RWSC and its expert Subcommittees, partners, and other participants to implement the Science Plan, including to understand the extent of ongoing and planned data collection activities, and to coordinate and plan future data collection and research activities with respect to offshore wind. The data will be displayed via online mapping platforms.

Extent

West -73.490000 East -73.030000
North 40.740000 South 40.400000

Scale Range

Maximum (zoomed in) 1:5,000
Minimum (zoomed out) 1:150,000,000

Topics and Keywords ▶

Themes or categories of the resource Biota, Oceans
Content type ⇔ Downloadable Data
Export to FGDC CSDGM XML format as Resource Description No

Citation ▶

Title Private Acoustic Telemetry Coverage Areas
Revision date 2026-06-23 00:00:00
Creation date 2025-03-03 00:00:00
Publication date 2025-04-04 00:00:00
Presentation formats ⇔ digital map

Citation Contacts ▶

Responsible party - originator
Individual's name Jordan Katz
Organization's name Regional Wildlife Science Collaborative

Resource Details ▶

Dataset languages ⇔ English (UNITED STATES)
Dataset character set utf8 - 8 bit UCS Transfer Format
Status on-going
Spatial representation type ⇔ vector

Supplemental information
This dataset displays bounding boxes depicting the locations of private acoustic telemetry receiver coverage areas submitted to the RWSC. The layer only includes projects that have receivers deployed beginning on 01/01/2022, including current projects and planned projects whose receivers have not yet been deployed.
Bounding boxes are provided by project personnel and plotted using the most recent deployment locations provided. The project/deployment start dates are also noted, and it can be assumed that receivers have been deployed, in some capacity, throughout the duration of the project. Please note that receiver deployments may change over the life of a project, and stations may go in and out of use. If you would like more specific information about a project's deployments, please reach out to the project personnel using the provided contact information.

Bounding boxes are symbolized by color according to the status of the project at the time of the last update. The color of the receiver deployments will not adjust to the time slider and will always depict the current project status.

Processing environment ⇌ Microsoft Windows 10 Version 10.0 (Build 26200) ; Esri ArcGIS 13.7.0.1901

Credits

RWSC Protected Fish Subcommittee: Jordan Katz, jordan.katz@noaa.gov

ArcGIS item properties

Name ⇌ rpt.rpt.Private_Acoustic_Telemetry_Receiver_Coverage_Areas

Location ⇌ Server=rwsc-db.env.duke.edu; Service=sde:postgresql:rWSC-db.env.duke.edu; Database=rpt; User=rpt; Version=sde.DEFAULT

Access protocol ⇌ ArcSDE Connection

Extents ►

Extent

Description

The temporal extent covers the current project deployment/start date.

Temporal extent

Beginning date 2009-01-01 00:00:00

Ending date 2025-03-31 00:00:00

Vertical extent

Minimum value ⇌ 0.000000

Maximum value ⇌ 0.000000

Extent

Geographic extent

Bounding rectangle

Extent type

Extent used for searching

West longitude ⇌ -73.490000

East longitude ⇌ -73.030000

North latitude ⇌ 40.740000

South latitude ⇌ 40.400000

Extent contains the resource ⇌ Yes

Extent in the item's coordinate system

westBL ⇌ -73.490000

eastBL ⇌ -73.030000

southBL ⇌ 40.400000

northBL ⇌ 40.740000

exTypeCode ⇌ Yes

Resource Maintenance ►

Resource maintenance

Update frequency as needed

Resource Constraints ►

Constraints

Limitations of use

These data will be used by RWSC and its expert Subcommittees, partners, and other participants to implement the Science Plan, including to understand the extent of ongoing and planned data collection activities, and to coordinate and plan future data collection and research activities with respect to offshore wind. The data will be displayed via online mapping platforms.

Spatial Reference ►

ArcGIS coordinate system

Type ⇌ Geographic

Geographic coordinate reference ⇌ GCS_WGS_1984

Coordinate reference details ⇌

GeographicCoordinateSystem

WKID 4326

XOrigin -400

YOrigin -400

XYScale 999999999.99999988

ZOrigin -100000

ZScale 10000

MOrigin -100000

MScale 10000

XYTolerance 8.983152841195215e-09

ZTolerance 0.001

MTolerance 0.001

HighPrecision true

LeftLongitude -180

LatestWKID 4326

WKT
GEOGCS["GCS_WGS_1984",DATUM["D_WGS_1984",SPHEROID["WGS_1984",6378137.0,298.257223563]],PRIMEM["Greenwich",0.0],UNIT["Degree",0.0174532925199433],AUTHORIT

Reference system identifier

Value ⇌ 4326
Codespace ⇌ EPSG
Version ⇌ 6.2(3.0.1)

Spatial Data Properties ►

Vector ►

Level of topology for this dataset ⇌ geometry only

Geometric objects

Feature class name rpt.rpt.Private_Acoustic_Telemetry_Receiver_Coverage_Areas
Object type ⇌ composite
Object count ⇌ 2

ArcGIS Feature Class Properties ►

Feature class name rpt.rpt.Private_Acoustic_Telemetry_Receiver_Coverage_Areas
Feature type ⇌ Simple
Geometry type ⇌ Polygon
Has topology ⇌ FALSE
Feature count ⇌ 2
Spatial index ⇌ TRUE
Linear referencing ⇌ FALSE

Data Quality ►

Data quality report - Conceptual consistency ►

Data quality measure reference

Measure description

The data are represented as bounding boxes. Device locations, deployment start/end dates and other metadata are subject to change. Reach out to the listed deployment POC for most up-to-date information.

Data quality report - Completeness omission ►

Data quality measure reference

Measure description

This dataset reflects the most recent present and future known locations of Acoustic Telemetry receivers, and is updated every four months. If an end date was not provided, an arbitrary end date was assigned to allow for time-enabled mapping feature to operate.

Lineage ►

Process step ►

When the process occurred 2025-01-01 00:00:00

Description

1. The coordinates of the bounding box are provided by the project personnel.
2. Column headings are set to match the other receiver layers.
3. All additional information is found on the project website or also provided by project personnel.

Process step ►

When the process occurred 2025-03-03 00:00:00

Description

Bounding Box brought into GIS using reference lat/long data

Process contact - processor

Individual's name Samantha Coccia-Schillo
Organization's name Regional Wildlife Science Collaborative
Contact's position GIS Manager

Contact information ►

Phone

Voice NA

Address

Type both

City NA

Administrative area NA

Postal code NA

Process step ►

When the process occurred 2025-04-04 00:00:00

Description

Layer published to server -service created.

Process contact - processor

Individual's name Samantha Coccia-Schillo
Organization's name Regional Wildlife Science Collaborative
Contact's position GIS Manager

Contact information ►

Phone

Voice NA

Address

Type both

City NA

Administrative area NA

Postal code NA

Distribution ►

Distribution format

Name ⇔ Enterprise Geodatabase Feature Class

Fields ►

Details for object rpt.rpt.Private_Acoustic_Telemetry_Receiver_Coverage_Areas ►

Type ⇔ Feature Class

Row count ⇔ 2

Field objectid ►

Alias ⇔ OBJECTID

Data type ⇔ OID

Width ⇔ 4

Precision ⇔ 10

Scale ⇔ 0

Field description ⇔

Internal feature number.

Description source ⇔

Esri

Description of values ⇔

Sequential unique whole numbers that are automatically generated.

Field shape ►

Alias ⇔ SHAPE

Data type ⇔ Geometry

Width ⇔ 8

Precision ⇔ 0

Scale ⇔ 0

Field description ⇔

Feature geometry.

Description source ⇔

Esri

Description of values ⇔

Coordinates defining the features.

Field project_name ►

Alias ⇔ project_name

Data type ⇔ String

Width ⇔ 255

Precision ⇔ 0

Scale ⇔ 0

Field project_description ►

Alias ⇔ project_description

Data type ⇔ String

Width ⇔ 255

Precision ⇔ 0

Scale ⇔ 0

Field collectioncode ►

Alias ⇔ collectioncode

Data type ⇔ String

Width ⇔ 255

Precision ⇔ 0

Scale ⇔ 0

Field affiliation ►

Alias ⇔ operators
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field contact_information ►

Alias ⇔ contact_information
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field approximation_technique ►

Alias ⇔ approximation_technique
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field station_name ►

Alias ⇔ station_name
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field projectdeployment_startdate ►

Alias ⇔ projectdeployment_startdate
Data type ⇔ Integer
Width ⇔ 4
Precision ⇔ 10
Scale ⇔ 0

Field projectrecovery_enddate ►

Alias ⇔ projectrecovery_enddate
Data type ⇔ Integer
Width ⇔ 4
Precision ⇔ 10
Scale ⇔ 0

Field last_deploy_date ►

Alias ⇔ last_deploy_date
Data type ⇔ Date
Width ⇔ 8
Precision ⇔ 0
Scale ⇔ 0

Field last_recovery_date ►

Alias ⇔ last_recovery_date
Data type ⇔ Date
Width ⇔ 8
Precision ⇔ 0
Scale ⇔ 0

Field station_type ►

Alias ⇔ station_type
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field instrument_type_model ►

Alias ⇔ instrument_type_model
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field co_deployed_instruments ►

Alias ⇔ co_deployed_instruments
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field co_deploy_list ►

Alias ⇔ co_deploy_list
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field archival_or_realtime_reciever ►

Alias ⇔ archival_or_realtime_reciever
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field rwsc_or_rosa_database ►

Alias ⇔ rwsc_or_rosa_database
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field regionalacoustic_telemtryntwrk ►

Alias ⇔ regionalacoustic_telemtryntwrk
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field project_status ►

Alias ⇔ project_status
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field receiver_status ►

Alias ⇔ receiver_status
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field station_status ►

Alias ⇔ station_status
Data type ⇔ String
Width ⇔ 255
Precision ⇔ 0
Scale ⇔ 0

Field seasonality_of_receivers ►

Alias ↔ seasonality_of_receivers
Data type ↔ String
Width ↔ 255
Precision ↔ 0
Scale ↔ 0

Field last_download_date ►

Alias ↔ last_download_date
Data type ↔ Date
Width ↔ 8
Precision ↔ 0
Scale ↔ 0

Field date_last_updated_by_rwsc ►

Alias ↔ date_last_updated_by_rwsc
Data type ↔ Date
Width ↔ 8
Precision ↔ 0
Scale ↔ 0

Field metadata_form ►

Alias ↔ metadata_form
Data type ↔ String
Width ↔ 255
Precision ↔ 0
Scale ↔ 0

Field affiliation_url ►

Alias ↔ affiliation_url
Data type ↔ String
Width ↔ 255
Precision ↔ 0
Scale ↔ 0

Field approximate_number_of_receivers ►

Alias ↔ approximate_number_of_receivers
Data type ↔ String
Width ↔ 255
Precision ↔ 0
Scale ↔ 0

Field network_link ►

Alias ↔ network_link
Data type ↔ String
Width ↔ 255
Precision ↔ 0
Scale ↔ 0

Field st_area(shape) ►

Alias ↔ st_area(shape)
Data type ↔ Double
Width ↔ 0
Precision ↔ 0
Scale ↔ 0

Field st_length(shape) ►

Alias ↔ st_length(shape)
Data type ↔ Double
Width ↔ 0
Precision ↔ 0
Scale ↔ 0

Metadata language ⇔ English (UNITED STATES)
Metadata character set utf8 - 8 bit UCS Transfer Format

Scope of the data described by the metadata dataset
Scope name ⇔ dataset

Last update ⇔ 2026-08-21

ArcGIS metadata properties
Metadata format ArcGIS 1.0
Standard or profile used to edit metadata FGDC

Created in ArcGIS for the item 2025-04-04 20:38:00
Last modified in ArcGIS for the item 2026-08-21 09:38:19

Automatic updates
Have been performed Yes
Last update 2026-08-21 09:38:19